「First degree.

Consider an equation $y' + a(t) \cdot y = 0$. Let's solve this.

If $y(t_0) = 0$ for some $t_0 \in I$, then since $\tilde{y} \equiv 0$ is a solution s.t. $\tilde{y}(t_0) = 0$, the uniqueness theorem gives $y \equiv 0$. Now, assume $y \neq 0$ for all $t \in I$.

Then, we obtain $\frac{y'}{y} = (\log y)' = -a(x)$, so

$$\log y = -\int_{t_0}^{t} a(x)\,dx \Rightarrow y = \exp\left(-\int_{t_0}^{t} a(x)\,dx\right)$$